

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\frac{dr}{dt} \propto \pm \frac{1}{r^2}$ or $\frac{dr}{dt} = \pm \frac{k}{r^2}$ (for k or a numerical k)	M1	3.3
	$\int r^2 dr = \int \pm k dt \Rightarrow \dots$ (for k or a numerical k)	M1	2.1
	$\frac{1}{3}r^3 = \pm kt \{+ c\}$	A1	1.1b
	$t = 0, r = 5$ and $t = 4, r = 3$ gives $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in minutes, is the time from when it {the mint} was placed in the mouth $t = 0, r = 5$ and $t = 240, r = 3$ gives $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, where r, in mm, is the radius {of the mint} and t, in seconds, is the time from when it {the mint} was placed in the mouth	M1	3.1a
		A1	1.1b
		(5)	
(b)	$r = 0 \Rightarrow 0 = -\frac{49}{6}t + \frac{125}{3} \Rightarrow 0 = -49t + 250 \Rightarrow t = \dots$	M1	3.4
	time = 5 minutes 6 seconds	A1	1.1b
		(2)	
(c)	Suggests a suitable limitation of the model. E.g. - Model does not consider how the mint is sucked - Model does not consider whether the mint is bitten - Model is limited for times up to 5 minutes 6 seconds, o.e. - Not valid for times greater than 5 minutes 6 seconds, o.e. - Mint may not retain the shape of a sphere (or have uniform radius) as it is being sucked - The model indicates that the radius of the mint is negative after it dissolves - Model does not consider the temperature in the mouth - Model does not consider rate of saliva production - Mint could be swallowed before it dissolves in the mouth	B1	3.5b
		(1)	
(8 marks)			

Notes for Question

(a)	
M1:	Translates the description of the model into mathematics. See scheme.
M1:	Separates the variables of their differential equation which is in the form $\frac{dr}{dt} = f(r)$ and some attempt at integration. (e.g. attempts to integrate at least one side). e.g. $\int r^2 dr = \int \pm k dt$ and some attempt at integration. Condone the lack of integral signs
Note:	You can imply the M1 mark for $r^2 dr = -k dt \Rightarrow \frac{1}{3}r^3 = -kt$
Note:	A numerical value of k (e.g. $k = \pm 1$) is allowed for the first two M marks
A1:	Correct integration to give $\frac{1}{3}r^3 = \pm kt$ with or without a constant of integration, c
M1:	For a complete process of using the boundary conditions to find both their unknown constants and finds an equation linking r and t So applies either $t = 0, r = 5$ and $t = 4, r = 3$, or $t = 0, r = 5$ and $t = 240, r = 3$, on their integrated equation to find their constants k and c and obtains an equation linking r and t
A1:	Correct equation, with variables r and t fully defined including correct reference to units. • $\frac{1}{3}r^3 = -\frac{49}{6}t + \frac{125}{3}$, {or an equivalent equation,} where r , in mm, is the radius {of the mint} and t , in minutes, is the time from when it {the mint} was placed in the mouth • $\frac{1}{3}r^3 = -\frac{49}{360}t + \frac{125}{3}$, {or an equivalent equation,} where r , in mm, is the radius {of the mint} and t , in seconds, is the time from when it {the mint} was placed in the mouth
Note:	Allow correct equations such as - in minutes, $r = \sqrt[3]{\frac{250-49t}{2}}$, $r^3 = -\frac{49}{2}t + 125$ or $t = \frac{250-2r^3}{49}$ - in seconds, $r = \sqrt[3]{\frac{15000-49t}{120}}$, $r^3 = -\frac{49}{120}t + 125$ or $t = \frac{15000-120r^3}{49}$
Note:	t defined as "the time from the start" is not sufficient for the final A1

(b)	
M1:	Sets $r = 0$ in their part (a) equation which links r with t and rearranges to make $t = \dots$
A1:	5 minutes 6 seconds cao (Note: 306 seconds with no reference to 5 minutes 6 seconds is A0)
Note:	Give M0 if their equation would solve to give a negative time or a negative time is found
Note:	You can mark part (a) and part (b) together
(c)	
B1:	See scheme
Note:	Do not accept by itself - mint may not dissolve at a constant rate - rate of decrease of mint must be constant $0 \leq t < \frac{250}{49}$, $r \geq 0$; without any written explanation - reference to a mint having $r > 5$

Q2.

Question	Scheme	Marks	AOs
(a)	Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$	B1	1.1a
	Substitutes either $P=0$ or $P=\frac{11}{2}$ into $1 = A(11-2P) + BP \Rightarrow A \text{ or } B$	M1	1.1b
	$\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$	A1	1.1b
	(3)		
(b)	Separates the variables $\int \frac{22}{P(11-2P)} dP = \int 1 dt$	B1	3.1a
	Uses (a) and attempts to integrate $\int \frac{2}{P} + \frac{4}{(11-2P)} dP = t + c$	M1	1.1b
	$2 \ln P - 2 \ln(11-2P) = t + c$	A1	1.1b
	Substitutes $t=0, P=1 \Rightarrow t=0, P=1 \Rightarrow c = (-2 \ln 9)$	M1	3.1a
	Substitutes $P=2 \Rightarrow t = 2 \ln 2 + 2 \ln 9 - 2 \ln 7$	M1	3.1a
	Time = 1.89 years	A1	3.2a
	(6)		

(c)	Uses ln laws $2 \ln P - 2 \ln(11-2P) = t - 2 \ln 9$ $\Rightarrow \ln\left(\frac{9P}{11-2P}\right) = \frac{1}{2}t$	M1	2.1
	Makes 'P' the subject $\Rightarrow \left(\frac{9P}{11-2P}\right) = e^{\frac{1}{2}t}$ $\Rightarrow 9P = (11-2P)e^{\frac{1}{2}t}$ $\Rightarrow P = f\left(e^{\frac{1}{2}t}\right) \text{ or } \Rightarrow P = f\left(e^{\frac{1}{2}t}\right)$	M1	2.1
	$\Rightarrow P = \frac{11}{2+9e^{\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$	A1	1.1b
	(3)		
(12 marks)			

Notes:

(a)

B1: Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$

M1: Substitutes $P=0$ or $P=\frac{11}{2}$ into $1=A(11-2P)+BP \Rightarrow A$ or B

Alternatively compares terms to set up and solve two simultaneous equations in A and B

A1: $\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$ or equivalent $\frac{1}{P(11-2P)} = \frac{1}{11P} + \frac{2}{11(11-2P)}$

Note: The correct answer with no working scores all three marks.

(b)

B1: Separates the variables to reach $\int \frac{22}{P(11-2P)} dP = \int 1 dt$ or equivalent

M1: Uses part (a) and $\int \frac{A}{P} + \frac{B}{(11-2P)} dP = A \ln P \pm C \ln(11-2P)$

A1: Integrates both sides to form a correct equation including a 'c' Eg
 $2 \ln P - 2 \ln(11-2P) = t + c$

M1: Substitutes $t=0$ and $P=1$ to find c

M1: Substitutes $P=2$ to find t . This is dependent upon having scored both previous M's

A1: Time = 1.89 years

(c)

M1: Uses correct log laws to move from $2 \ln P - 2 \ln(11-2P) = t + c$ to $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d$ for their numerical 'c'

M1: Uses a correct method to get P in terms of $e^{\frac{1}{2}t}$

This can be achieved from $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d \Rightarrow \frac{P}{11-2P} = e^{\frac{1}{2}t+d}$ followed by cross multiplication and collection of terms in P (See scheme)

Alternatively uses a correct method to get P in terms of $e^{-\frac{1}{2}t}$. For example

$\frac{P}{11-2P} = e^{\frac{1}{2}t+d} \Rightarrow \frac{11-2P}{P} = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \frac{11}{P} - 2 = e^{-\left(\frac{1}{2}t+d\right)} \Rightarrow \frac{11}{P} = 2 + e^{-\left(\frac{1}{2}t+d\right)}$ followed by division

A1: Achieves the correct answer in the form required. $P = \frac{11}{2 + 9e^{-\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$ oe

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$dh = -2(4 - u) du$	B1	This mark is given for finding an expression for dh
	$\int \frac{dh}{4 - \sqrt{h}} = \int \frac{-2(4 - u) du}{4 - \sqrt{h}}$	M1	This mark is given for substituting $u = 4 - \sqrt{h}$ into the integral
	$= \int -\frac{8}{u} + 2 du$	M1	This mark is given for a method to find a simplified version of the integral
	$-8 \ln u + 2u + c$ $= -8 \ln(4 - \sqrt{h}) + 2(4 - \sqrt{h}) + c$	M1	This mark is given for integrating with respect to u to produce an expression in terms of h
		A1	This mark is given for a correct expression for the integral
	$= -8 \ln(4 - \sqrt{h}) - 2\sqrt{h} + k$	A1	This mark is given for a full proof to arrive at the answer as shown (appreciating that $k = c + 8$)
(b)	$\frac{dh}{dt} = 0 \Rightarrow 4 - \sqrt{h} = 0$	M1	This mark is given for a setting $\frac{dh}{dt} = 0$
	$0 < h < 16$	A1	This mark is given for deducing the range of the heights of the trees for this model
(c)	$\frac{dh}{dt} = \frac{t^{0.25}(4 - \sqrt{h})}{20} \Rightarrow \frac{dh}{(4 - \sqrt{h})} = \frac{t^{0.25} dt}{20}$	B1	This mark is given for separating the variables
	$-8 \ln(4 - \sqrt{h}) - 2\sqrt{h} + k = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to integrate both sides of the equation
		A1	This mark is given for integrating both sides of the equation correctly
	When $t = 0$ and $h = 1$, $-8 \ln 3 - 2 + k = 0$ $k = 2 + 8 \ln 3$	M1	This mark is given for substituting values of $t = 0$ and $h = 1$ to find a value for k
	When $h = 12$, $-8 \ln(4 - \sqrt{12}) - 2\sqrt{12} + 2 + 8 \ln 3 = \frac{t^{1.25}}{25}$	M1	This mark is given for a method to find a value for t by substituting $h = 12$ into the equation
	$t^{1.25} = 221.2795 \Rightarrow t = \sqrt[1.25]{221.2795}$	M1	This mark is given for simplifying to find an expression for t
	$t = 75.2$ years	A1	This mark is given for correctly finding the time the tree would take to reach a height of 12 metres
(Total 15 marks)			

Q4.

Question	Scheme	Marks	AOs
(a)	$x^2 + 8x - 3 = (Ax + B)(x + 2) + C \text{ or } Ax(x + 2) + B(x + 2) + C$ $\Rightarrow A = \dots, B = \dots, C = \dots$ or $ \begin{array}{r} x+6 \\ x+2 \overline{) x^2 + 8x - 3} \\ \underline{x^2 + 2x} \\ 6x - 3 \\ \underline{6x + 12} \\ -15 \end{array} $	M1	1.1b
	Two of $A = 1, B = 6, C = -15$	A1	1.1b
	All three of $A = 1, B = 6, C = -15$	A1	1.1b
		(3)	
6(b)	$\int \frac{x^2 + 8x - 3}{x + 2} dx = \int x + 6 - \frac{15}{x + 2} dx = \dots - 15 \ln(x + 2)$	M1	1.1b
	$= \frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \quad (+c)$	A1ft	1.1b
	$ \begin{aligned} \int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx &= \left[\frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \right]_0^6 \\ &= (18 + 36 - 15 \ln 8) - (0 + 0 - 15 \ln 2) \\ &= 18 + 36 - (15 - 45) \ln 2 \text{ or e.g. } 18 + 36 + 15 \ln \left(\frac{2}{8} \right) \end{aligned} $	M1	2.1
	$= 54 - 30 \ln 2$	A1	1.1b
		(4)	
(7 marks)			

Notes:

(a)

M1: Multiplies by $(x + 2)$ and attempts to find values for A, B and C e.g. by comparing coefficients or substituting values for x . If the method is unclear, at least 2 terms must be correct on rhs.

Or attempts to divide $x^2 + 8x - 3$ by $x + 2$ and obtains a linear quotient and a constant remainder.

This mark may be implied by 2 correct values for A, B or C

A1: Two of $A = 1, B = 6, C = -15$. But note that just performing the division correctly is insufficient and they must clearly identify their A, B, C to score any accuracy marks.

A1: All three of $A = 1, B = 6, C = -15$

This is implied by stating $\frac{x^2 + 8x - 3}{x + 2} = x + 6 - \frac{15}{x + 2}$ or within the integral in (b)

(b)

M1: Integrates an expression of the form $\frac{C}{x + 2}$ to obtain $k \ln(x + 2)$.

Condone the omission of brackets around the " $x + 2$ "

A1ft: Correct integration ft on their $Ax + B + \frac{C}{x + 2}$, ($A, B, C \neq 0$) The brackets should be present around the " $x + 2$ "

unless they are implied by subsequent work.

M1: Substitutes both limits 0 and 6 into an expression that contains an x or x^2 term or both and a $\ln$ term and subtracts either way round **WITH** fully correct log work to combine two log terms (but allow sign errors when removing brackets) leading to an answer of the form $a + b \ln c$ (a, b and c not necessarily integers)

e.g. if they expand to get $-15 \ln 8 - 15 \ln 2$ followed by $-15 \ln 16$ and reach $a + b \ln c$ then allow the M mark

A1: $54 - 30 \ln 2$ (Apply isw once a correct answer is seen)



Q5.

Question	Scheme	Marks	AOs
(a)	$\frac{dV}{dt} = 0.48 - 0.1h$	B1	3.1b
	$V = 24h \Rightarrow \frac{dV}{dh} = 24 \text{ or } \frac{dh}{dV} = \frac{1}{24}$	B1	3.1b
	$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV} = \frac{0.48 - 0.1h}{24}$ or e.g. $\frac{dV}{dt} = \frac{dV}{dh} \frac{dh}{dt} \Rightarrow 0.48 - 0.1h = 24 \frac{dh}{dt}$	M1	2.1
	$1200 \frac{dh}{dt} = 24 - 5h^*$	A1*	1.1b
	(4)		
(b)	$1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \int \frac{1200}{24 - 5h} dh = \int dt$ $\Rightarrow$ e.g. $\alpha \ln(24 - 5h) = t(+c)$ oe or $1200 \frac{dh}{dt} = 24 - 5h \Rightarrow \frac{dt}{dh} = \frac{1200}{24 - 5h}$ $\Rightarrow$ e.g. $t(+c) = \alpha \ln(24 - 5h)$ oe	M1	3.1a
	$t = -240 \ln(24 - 5h)(+c)$ oe	A1	1.1b
	$t = 0, h = 2 \Rightarrow 0 = -240 \ln(24 - 10) + c \Rightarrow c = \dots (240 \ln 14)$	M1	3.4
	$t = 240 \ln(14) - 240 \ln(24 - 5h)$	A1	1.1b
	$t = 240 \ln \frac{14}{24 - 5h} \Rightarrow \frac{t}{240} = \ln \frac{14}{24 - 5h} \Rightarrow e^{\frac{t}{240}} = \frac{14}{24 - 5h}$ $\Rightarrow 14e^{-\frac{t}{240}} = 24 - 5h \Rightarrow h = \dots$	ddM1	2.1
	$h = 4.8 - 2.8e^{-\frac{t}{240}}$ oe e.g. $h = \frac{24}{5} - \frac{14}{5}e^{-\frac{t}{240}}$	A1	3.3
	(6)		

(c)	Examples: - As $t \rightarrow \infty, e^{-\frac{t}{240}} \rightarrow 0$ - When $h > 4.8, \frac{dV}{dt} < 0$ - Flow in = flow out at max h so $0.1h = 4.8 \rightarrow h = 4.8$ - As $e^{-\frac{t}{240}} > 0, h < 4.8$ $h = 5 \Rightarrow \frac{dV}{dt} = -0.02$ or $\frac{dh}{dt} = -\frac{1}{1200}$ $\frac{dh}{dt} = 0 \Rightarrow h = 4.8$ $h = 5 \Rightarrow 4.8 - 2.8e^{-\frac{t}{240}} = 5 \Rightarrow e^{-\frac{t}{240}} < 0$	M1	3.1b
	- The limit for h (according to the model) is 4.8m and the tank is 5m high so the tank will never become full - If $h = 5$ the tank would be emptying so can never be full - The equation can't be solved when $h = 5$	A1	3.2a
	(2)		

Notes
(a) B1: Identifies the correct expression for $\frac{dV}{dt}$ according to the model B1: Identifies the correct expression for $\frac{dV}{dh}$ according to the model M1: Applies $\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$ or equivalent correct formula with their $\frac{dV}{dt}$ and $\frac{dV}{dh}$ which may be implied by their working A1*: Correct equation obtained with no errors Note that: $\frac{dV}{dt} = 0.48 - 0.1h \Rightarrow \frac{dh}{dt} = \frac{0.48 - 0.1h}{24} \Rightarrow 1200 \frac{dh}{dt} = 24 - 5h$ * scores B1B0M0A0. There must be clear evidence where the “24” comes from and evidence of the correct chain rule being applied.
(b) M1: Adopts a correct strategy by separating the variables correctly or rearranges to obtain $\frac{dt}{dh}$ correctly in terms of h and integrates to obtain $t = \alpha \ln(24 - 5h) (+c)$ or equivalent (condone missing brackets around the “24 – 5h”) and + c not required for this mark. A1: Correct equation in any form and + c not required. Do not condone missing brackets unless they are implied by subsequent work. M1: Substitutes $t = 0$ and $h = 2$ to find their constant of integration (there must have been some attempt to integrate) A1: Correct equation in any form ddM1: Uses fully correct log work to obtain h in terms of t. This depends on both previous method marks. A1: Correct equation Note that the marks may be earned in a different order e.g.: $t + c = -240 \ln(24 - 5h) \Rightarrow -\frac{t}{240} + d = \ln(24 - 5h) \Rightarrow Ae^{-\frac{t}{240}} = 24 - 5h$ $t = 0, h = 2 \Rightarrow A = 14 \Rightarrow 14e^{-\frac{t}{240}} = 24 - 5h \Rightarrow h = 4.8 - 2.8e^{-\frac{t}{240}}$ Score as M1 A1 as in main scheme then M1: Correct work leading to $Ae^{\frac{t}{240}} = 24 - 5h$ (must have a constant “A”) $A1: Ae^{-\frac{t}{240}} = 24 - 5h$ ddM1: Uses $t = 0, h = 2$ in an expression of the form above to find A $A1: h = 4.8 - 2.8e^{-\frac{t}{240}}$ (c) M1: See scheme for some examples A1: Makes a correct interpretation for their method. There must be no incorrect working or contradictory statements. This is not a follow through mark and if their equation in (b) is used it must be correct.

Q6.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t$ or $5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2

	$(\text{Area}) = \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area}) = \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt \quad *$	A1*	2.1*
		(3)	
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = \left[-20 \cos^3 t \right]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 \quad *$	A1*	2.1
		(3)	
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
		(5)	
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $\int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “ dt ”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = \dots$

A1: At least one correct value for t , correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form A1: Correct quartic M1: Solves quartic A1: Correct values

A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. Units are required.

Q7.

Question	Scheme	Marks	AOs
(a)	$\frac{dH}{dt} = \frac{H \cos 0.25t}{40} \Rightarrow \int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$	M1	3.1a
	$\ln H = \frac{1}{10} \sin 0.25t (+c)$	M1	1.1b
		A1	1.1b
	Substitutes $t = 0, H = 5 \Rightarrow c = \ln(5)$	dM1	3.4
	$\ln\left(\frac{H}{5}\right) = \frac{1}{10} \sin 0.25t \Rightarrow H = 5e^{0.1 \sin 0.25t} *$	A1*	2.1
		(5)	
(b)	Max height = $5e^{0.1} = 5.53 \text{ m}$ (Condone lack of units)	B1	3.4
		(1)	
(c)	Sets $0.25t = \frac{5\pi}{2}$	M1	3.1b
	31.4	A1	1.1b
		(2)	
(8 marks)			

(a)

M1: Separates the variables to reach $\int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$ or equivalent.

The integral signs need to be present on both sides and the dH AND dt need to be in the correct positions.

M1: Integrates both sides to reach $\ln H = A \sin 0.25t$ or equivalent with or without the $+ c$

A1: $\ln H = \frac{1}{10} \sin 0.25t + c$ or equivalent with or without the $+ c$. Allow two constants, one either side

If the 40 was on the lhs look for $40 \ln H = 4 \sin 0.25t + c$ or equivalent.

dM1: Substitutes $t = 0, H = 5 \Rightarrow c = \dots$ There needs to have been a single $" + c "$ to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see $"t = 0, H = 5"$ as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg. $40 \ln H = 4 \sin 0.25t + c \Rightarrow H^{40} = e^{4 \sin 0.25t} + e^c \Rightarrow 5^{40} = 1 + e^c \Rightarrow c = \dots$

Also many students will be attempting to get to the given answer so condone the method of finding $c = \dots$. These students will lose the A1* mark

A1*: Proceeds via $\ln H = \frac{1}{10} \sin 0.25t + \ln 5$ or equivalent to the given answer $H = 5e^{0.1 \sin 0.25t}$ with at least one correct intermediate line and no incorrect work.

DO NOT condone c 's going to c 's when they should be e^c or A

Accept as a minimum $\ln H = \frac{1}{10} \sin 0.25t + \ln 5 \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + \ln 5}$ or $H = e^{\frac{1}{10} \sin 0.25t} \times e^{\ln 5}$ before sight of the given answer

If the only error was to omit the integration signs on line 1, thus losing the first M1, allow the candidate to have access to this mark following a correct intermediate line (see above).

If they attempt to change the subject first then the constant of integration must have been adapted if the A1* is to be awarded. $\ln H = \frac{1}{10} \sin 0.25t + c \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + c} \Rightarrow H = Ae^{\frac{1}{10} \sin 0.25t}$

The dM1 and A1* under this method are awarded at virtually the same time.

Also, for the final two marks, you may see a proof from $\int_0^H \frac{40}{H} dH = \int_0^t \cos 0.25t dt$

.....
There is an alternative via the use of an integrating factor.

(b)

B1: States that the maximum height is 5.53 m Accept $5e^{0.1}$ Condone a lack of units here, but penalise if incorrect units are used.

(c)

M1: For identifying that it would reach the maximum height for the 2nd time when $0.25t = \frac{5\pi}{2}$ or 450

A1: Accept awrt 31.4 or 10π Allow if units are seen

Q8.

Question	Scheme	Marks	AOs
(a)	$x = u^2 + 1 \Rightarrow dx = 2u du$ oe	B1	1.1b
	Full substitution $\int \frac{3dx}{(x-1)(3+2\sqrt{x-1})} = \int \frac{3 \times 2u du}{(u^2+1-1)(3+2u)}$	M1	1.1b
	Finds correct limits e.g. $p = 2, q = 3$	B1	1.1b
	$= \int \frac{3 \times 2u du}{u^2(3+2u)} = \int \frac{6 du}{u(3+2u)} *$	A1*	2.1
		(4)	
(b)	$\frac{6}{u(3+2u)} = \frac{A}{u} + \frac{B}{3+2u} \Rightarrow A = \dots, B = \dots$	M1	1.1b
	Correct PF: $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$	A1	1.1b
	$\int \frac{6 du}{u(3+2u)} = 2 \ln u - 2 \ln(3+2u) \quad (+c)$	dM1 A1ft	3.1a 1.1b
	Uses limits $u = "3", u = "2"$ with some correct $\ln$ work leading to $k \ln b$ E.g. $(2 \ln 3 - 2 \ln 9) - (2 \ln 2 - 2 \ln 7) = 2 \ln \frac{7}{6}$	M1	1.1b
	$\ln \frac{49}{36}$	A1	2.1
		(6)	
(10 marks)			
Notes: Mark (a) and (b) together as one complete question			

- (a)
- B1:** $dx = 2u du$ or exact equivalent. E.g. $\frac{dx}{du} = 2u, \frac{du}{dx} = \frac{1}{2}(x-1)^{-\frac{1}{2}}$
- M1:** Attempts a full substitution of $x = u^2 + 1$, including $dx \rightarrow \dots du$ to form an integrand in terms of u .
 Condone slips but there should be an attempt to use the correct substitution on the denominator.
- B1:** Finds correct limits either states $p = 2, q = 3$ or sight of embedded values as limits to the integral
- A1*:** Clear reasoning including one fully correct intermediate line, including the integral signs, leading to the given expression ignoring limits. So B1, M1, B0, A1 is possible if the limits are incorrect, omitted or left as 5 and 10.
- (b)
- M1:** Uses correct form of PF leading to values of A and B .
- A1:** Correct PF $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$ (Not scored for just the correct values of A and B)
- dM1:** This is an overall problem solving mark. It is for using the correct PF form and integrating using $\ln$ s.
 Look for $P \ln u + Q \ln(3+2u)$
- A1ft:** Correct integration for their $\frac{A}{u} + \frac{B}{3+2u} \rightarrow A \ln u + \frac{B}{2} \ln(3+2u)$ with or without modulus signs
- M1:** Uses their 2 and 3 as limits, with at least one correct application of the addition law or subtraction law leading to the form $k \ln b$ or $\ln a$. PF's must have been attempted. Condone bracketing slips. Alternatively changing the u 's back to x 's and use limits of 5 and 10.
- A1:** Proceeds to $\ln \frac{49}{36}$. Answers without working please send to review.